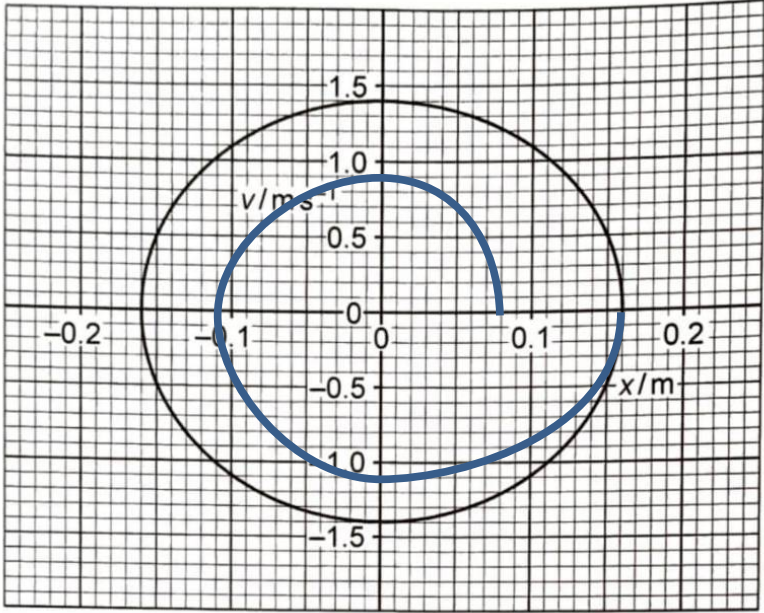


Qn	Solution
1(a)	$T = 2\pi\sqrt{\frac{m}{k}}$ $k = \frac{4\pi^2 m}{T^2} = \frac{4\pi^2(0.120)}{(7.2/10)^2} = 9.1385 \text{ N m}^{-1} \text{ (5 s.f.)}$ $\frac{\Delta k}{k} = \frac{\Delta m}{m} + 2\frac{\Delta T}{T}$ $= 0.01 + 2\frac{(0.2/10)}{(7.2/10)}$ $= 0.0656$ $\Delta k = 0.0656 \times 9.1385 = 0.599 \approx 0.6 \text{ (1 s.f.)}$ $\therefore k = 9.1 \pm 0.6 \text{ N m}^{-1}$
1(b)(i)	$a_{\max} = \omega^2 x_0$ $= \left(\frac{2\pi}{T}\right)^2 x_0$ $= \left(\frac{2\pi}{0.72}\right)^2 (0.16)$ $= 12.2 \text{ m s}^{-2}$ $= 12 \text{ m s}^{-2}$
1(b)(ii)	 Fig. 1.2

2(a)(i)	A vector quantity is one that has magnitude and direction
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